Curriculum Vitae

Woorak Choi

Postdoctoral Researcher, McMaster University

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Research Goal: *To understand how the interstellar medium evolves across scales from giant molecular clouds to sub-kpc, and how environmental processes/mechanisms (e.g. bars, ram pressure stripping, mergers, magnetic fields) affect this evolution in nearby galaxies, using multi-wavelength observations (ALMA, JWST, VLA) combined with high-resolution numerical simulations.*

Education

Ph.D. in Astronomy — Yonsei University, Seoul, Korea 2017.09 – 2024.02

Thesis: "High-resolution radio observations and GMC-scale simulations as probes of galaxy evolution"

Advisor: Prof. Aeree Chung

B.Sc. in Astronomy — Yonsei University, Seoul, Korea 2013.03 – 2017.08

Previous and Current Positions

Postdoctoral Researcher — McMaster University 2024.09 –

Working with Prof. Christine D. Wilson on JWST, VLA and ALMA observations of nearby galaxies.

- Leading a JWST analysis of NGC 4826 (M64) covering PAH emission, Pa-α/Br-α recombination line maps, and their correlation with CO(3–2) at ∼8 pc resolution (Choi et al., in prep., with J. Sun & C. Wilson).
- Leading the study of giant molecular clouds in Virgo galaxies using MAUVE-ALMA data (Choi et al., in prep., with J. Sun & C. Wilson).
- One successful VLA-JWST joint proposal (GO 9246): targeting 5 nearby nuclear-ring host galaxies.
- Three successful VLA proposals: two for targeting 12 Virgo cluster galaxies to study radio continuum and magnetic field, and the other for targeting 11 nearby nuclear rings to study radio continuum and magnetic field.
- Leading two ALMA proposals: one targeting 21 barred galaxies CO(2–1) at ∼200 pc resolution, and another observing nuclear ring GMCs in 8 targets at ∼15 pc.
- Active member of the MAUVE and KILOGAS collaborations.

Postdoctoral Researcher — Yonsei University 2024.03 – 2024.06

Continued work with Prof. Aeree Chung on high-resolution VLA continuum observations of the ram-pressure-stripped Virgo galaxy NGC 4522, characterising the radio spectral index and magnetic field structure (Choi et al., 2026).

Visiting Researcher — University of Oxford 2019.11 – 2020.02

Worked with Prof. Martin Bureau on ALMA molecular gas observations of nearby early-type galaxies as part of

the WISDOM collaboration. This visit initiated my long-running involvement with WISDOM, leading to multiple first-author and co-author GMC papers.

Research Assistant — Yonsei University 2017.09 – 2024.02
PhD research with Prof. Aeree Chung on (i) high-resolution ALMA CO observations of GMCs in barred spiral galaxies and (ii) TIGRESS-based GMC-scale simulations of ram pressure stripping of the multiphase ISM.

Research Experience

Molecular Gas and Giant Molecular Clouds in Nearby Galaxies

My central research interest is the characterisation of molecular gas and GMC properties in nearby galaxies, and how galactic environment regulates star formation at the cloud scale. I have led GMC studies in the central regions of barred spiral galaxies as part of the WISDOM collaboration (Choi et al., 2023 — NGC 5806; Choi et al., 2024 — NGC 613). I am now extending this work to other nearby galaxies using ALMA archival data (Choi et al., in prep), Virgo cluster galaxies through MAUVE-ALMA (Choi et al., in prep.) and to simulated nuclear rings (Choi et al., in prep., with S. Moon, C.-G. Kim, A. Chung).

JWST Data Analysis and Pipeline Development

I lead the JWST analysis of M64 (NGC 4826) using NIRCam and MIRI imaging, producing PAH emission maps and Pa-α/Br-α recombination line maps and correlating them with ALMA CO(3–2) at ∼8 pc physical resolution. In parallel, I am developing a production-quality Python pipeline for JWST emission line map generation. I am co-PI of the joint VLA/JWST program GO 9246, supported by a CSA grant (CAD 60,000).

Ram Pressure Stripping: Observations and Simulations

I have studied environmental gas-stripping from both theoretical and observational sides. Observationally, I led a high-resolution VLA radio continuum study of NGC 4522, constraining the radio spectral index and magnetic field structure (Choi et al., 2026). New radio continuum data for Virgo galaxies will be obtained through multiple upcoming observations. Using the TIGRESS framework, I investigated ram pressure stripping of the multiphase ISM at GMC-scale, quantifying how cold, warm, and hot gas respond to varying ICM conditions (Choi, Kim & Chung, 2022). I am now extending this work to galaxy-wide simulations (with T. Kimm, J. Lee, A. Chung).

Radio Interferometry

I am highly experienced in ALMA and VLA data reduction in CASA, including multi-configuration combination, self-calibration, and continuum/line imaging. I have led successful ALMA and VLA proposals as PI or co-PI in multiple cycles (see the Successful Telescope Proposals section), and I organised an ALMA proposal preparation workshop at McMaster (2026.03). Using VLA observations, I have developed expertise in the analysis of polarised radio continuum data, particularly for inferring magnetic field strength and topology in environmentally affected cluster galaxies (Choi et al., 2026).

Numerical Simulations of the ISM

Through my PhD I developed working expertise with the TIGRESS framework (Athena-based) for high-resolution (GMC scale) simulations of the stratified, multiphase ISM. I have ongoing experience with RAMSES and Athena, and apply these tools to nuclear ring and ram-pressure environments.

Collaborations

I am an active member of three major survey collaborations: WISDOM (CO mapping of early-type galaxies), MAUVE (multi-phase studies of Virgo cluster galaxies with MUSE + ALMA), and KILOGAS (kiloparsec-scale CO

mapping of nearby galaxies). I maintain shared data analysis tools and pipelines for these collaborations.

Successful Telescope Proposals

I have a strong record of obtaining observation time on leading radio and infrared facilities, with awarded time on ALMA, JWST, VLA, and MeerKAT through PI, co-PI, and co-I roles across multiple cycles.

As PI / co-PI

- VLA A & B-configurations, 27A-488, 69.7 hours (PI)
- VLA C-configuration, 26B-220, 20.4 hours, Priority B (PI)
- VLA D-configuration, 26A-474, 53.5 hours, Priority C (PI)
- VLA C-configuration, 20A-310, 7.4 hours, Priority B (PI)
- ALMA Cycle 8, 2021.1.01491.S, 15.3 hours (PI)
- VLA B-configuration, 25B-266, 30 hours (co-PI; PI: Anan Lu)
- JWST GO 9246, 13 hours, joint VLA/JWST (co-PI)

As co-I

- ALMA Cycle 13, 2026.1.01665.S (PI: B. Lee)
- ALMA Cycle 13, 2026.1.00580.S (PI: T. Davis)
- ALMA Cycle 13, 2026.1.00078.S (PI: T. Williams)
- ALMA Cycle 12, 2025.1.00193.S (PI: J. Sun)
- ALMA Cycle 11, 2024.1.00297.S (PI: T. Williams)
- ALMA Cycle 9, 2022.1.00141.S (PI: T. Williams)
- ALMA Cycle 9, 2022.1.01122.S (PI: M. Bureau)
- ALMA Cycle 9, 2022.1.01556.S (PI: B. Lee)
- ALMA Cycle 8, 2021.1.01098.S (PI: L. Liu)
- VLA B-configuration, 24A-294, 22 hours (PI: A. Lu)
- VLA B-configuration, 23A-265, 5.5 hours (PI: A. Lu)
- MeerKAT S-band, MKT-25083, 38 hours, Priority B1 (PI: I. Roberts)

Awards & Fellowships

- CSA grant funding for JWST program GO 9246 (CAD 60,000, 2025–2027)
- NAASC ALMA Ambassador, Cycle 13 (USD 5,000 travel funding, 2026.02 – 2027.02)
- Postdoctoral Fellowship, NRF of Korea — Grant No. RS-2024-00413394 (USD 43,000, 2024.10–2025.09)
- Brain Korea 21 (BK21) scholarship (2017.09–2023.08)

Publication Record

I have a steadily growing publication record in leading astronomy journals, with 4 first-author refereed papers, 12 refereed co-author papers, and multiple papers in preparation (including 3 as first author).

First-author refereed papers

Choi, W., et al. (2026), "The impact of ram pressure on the radio spectral index and magnetic field of NGC 4522: A high-resolution VLA continuum study," ApJ. DOI: https://doi.org/10.3847/1538-4357/ae39ca

Choi, W., et al. (2024), "WISDOM Project — XXI. Giant Molecular Clouds in the Central Region of the Barred Spiral Galaxy NGC 613," MNRAS. DOI: https://doi.org/10.1093/mnras/stae1394

Choi, W., et al. (2023), "WISDOM Project — XV. Giant Molecular Clouds in the Central Region of the Barred Spiral Galaxy NGC 5806," MNRAS. DOI: https://doi.org/10.1093/mnras/stad1211

Choi, W., Kim, C.-G., & Chung, A. (2022), "Ram Pressure Stripping of the Multiphase ISM: A Detailed View from TIGRESS Simulations," ApJ. DOI: https://doi.org/10.3847/1538-4357/ac82ba

First-author papers in preparation

Choi, W., Sun, J., Wilson, C., "PAHs and star-forming regions of M64 using JWST NIRCam/MIRI," in preparation.

Choi, W., Sun, J., MAUVE team, "MAUVE-ALMA: Giant Molecular Clouds of Virgo cluster galaxies," in preparation.

Choi, W., Moon, S., Kim, C.-G., Chung, A., "Structure and kinematics of molecular gas at the cloud scale in the simulated nuclear ring," in preparation.

Co-author refereed papers

Pan, H.-A., ... Choi, W., et al. (2026), "Tracing Radial Structures of Molecular Gas and Gas-phase Metallicity Across the Galaxy Merger Sequence with KILOGAS", ApJ.

Brown, T., … Choi, W., et al. (2026), "MAUVE–MUSE: Ionization and Kinematic Signatures of Environmental Effects on Virgo Cluster Disks," ApJ. DOI: https://doi.org/10.3847/1538-4357/ae632e

Liang, F.-H., … Choi, W., et al. (2026), "WISDOM Project — XXVII. Giant molecular clouds of the lenticular galaxy NGC 1387," MNRAS. DOI: https://doi.org/10.1093/mnras/stag221

Williams, T. G., … Choi, W., et al. (2025), "The resolved star-formation efficiency of early-type galaxies," MNRAS. DOI: https://doi.org/10.1093/mnras/staf498

Lu, A., … Choi, W., et al. (2025), "WISDOM Project — XXIII. Star-formation efficiencies of eight early-type galaxies and bulges," MNRAS. DOI: https://doi.org/10.1093/mnras/staf675

Liu, L., … Choi, W., et al. (2025), "WISDOM Project — XXIV. Giant molecular clouds of the spiral galaxy NGC 5064," MNRAS. DOI: https://doi.org/10.1093/mnras/staf1159

Barbara, C., … Choi, W., et al. (2025), "Multiphase Astrophysics to Unveil the Virgo Environment (MAUVE)," The Messenger. DOI: https://doi.org/10.18727/0722-6691/5393

Lu, A., … Choi, W., et al. (2024), "WISDOM Project — XX. Strong shear tearing molecular clouds apart in NGC 524," MNRAS. DOI: https://doi.org/10.1093/mnras/stae1395

Williams, T. G., … Choi, W., et al. (2023), "WISDOM Project — XVII. Beam-by-beam Properties of the Molecular Gas in Early-type Galaxies," MNRAS. DOI: https://doi.org/10.1093/mnras/stad2455

Liu, L., … Choi, W., et al. (2022), "WISDOM Project — XII. Clump properties and turbulence regulated by clump-clump collisions in NGC 404," MNRAS. DOI: https://doi.org/10.1093/mnras/stac2287

Lu, A., … Choi, W., et al. (2022), "WISDOM Project — XI. Star formation efficiency in the bulge of the AGN-host

galaxy NGC 3169," MNRAS. DOI: https://doi.org/10.1093/mnras/stac1583

Davis, T. A., … Choi, W., et al. (2022), "WISDOM Project — X. The morphology of the molecular ISM in galaxy centres," MNRAS. DOI: https://doi.org/10.1093/mnras/stac600

Co-author papers in preparation

Dominiak, P., … Choi, W., et al., "WISDOM Project — Catalogue of CO observations of nearby galaxies and the CO Tully-Fisher relation," submitted to MNRAS

Konishi, A., … Choi, W., et al., "WISDOM Project — Giant Molecular Clouds at the central region of NGC 1574," submitted to MNRAS

Ledger, B., … Choi, W., et al., "KILOGAS: A centrally peaked enhancement of molecular gas in post-merger galaxies", submitted to MNRAS

Davis, T., … Choi, W., et al., "KILOGAS -- A legacy survey for the resolved extragalactic molecular interstellar medium: Survey overview and the molecular mass -- size relationship", in prep.

Teaching, Supervision, and Service

Supervision

- Masha Kilibarda – Master Student, McMaster University (co-supervising, 2026.09 – present)
- Evan Barry — Summer undergraduate intern, University of British Columbia (co-supervising, 2026.05 – 08)
- Jennifer Laing — Ph. D student, McMaster University (co-supervising, 2025.01 – present)
- Babara Meave — Ph. D student, McMaster University (co-supervising, 2025.01 – present)
- Jose Carlos — Mitacs summer undergraduate intern, McMaster University (2025.05 – 08)

Teaching

- Graduate course (Interstellar Medium), McMaster University (scheduled, 2027.03-2027.04)
- Teaching Assistant, Yonsei University: Introduction to Astronomy I (2019-1); Introduction to Astronomy II (2018-2); Astrophysics I (2018-1); Astronomy 101 (2017-2)
- Teaching Assistant, advanced course for middle-school students, Yonsei University (2016.02–2017.02)

Service and Community

- ApJ referee and ALMA proposal reviewer (2024 – present)
- NRAO Science Review Panel (2026.07 – 2028.07)
- JWST Cycle 6 Telescope Allocation Committee (2026.09 - 2027.03)
- President, Korea Young Astronomers Meeting (KYAM, official division of the Korean Astronomical Society) (2022.03–2023.02)
- ALMA proposal preparation workshop, McMaster University (2026.03, as ALMA Ambassador, Cycle 13)
- Volunteer, IAU General Assembly Busan (40 hours, 2022)

Presentations & Posters

Selected oral talks (O) and posters (P) at international and domestic meetings and seminars.

2026.06 **O** Canadian Astronomical Society's Annual General Meeting *Montreal*

2026.04	**O**	Astro Colloquium, Sejong University	*Seoul*
2026.04	**O**	Meeting of the Korean Astronomical Society	*Suwon*
2026.04	**O**	Astro Colloquium, Seoul National University	*Seoul*
2026.04	**O**	Galaxy Forum, Yonsei University	*Seoul*
2026.03	**O**	ALMA proposal preparation workshop	*Hamilton*
2025.10	**O**	GEESE-ON workshop	*Waterloo*
2025.10	**O**	Astro Seminar, University of Waterloo	*Waterloo*
2025.05	**P**	The Ecology of Galaxies	*Hungary*
2024.12	**O**	Astronomy Journal Club, McMaster University	*Hamilton*
2024.04	**O**	Center for Galaxy Evolution Research Forum	*Seoul*
2024.01	**O**	East Asia Young Astronomers Meeting	*Thailand*
2024.01	**O**	243rd American Astronomical Society Meeting	*USA*
2023.04	**O**	Meeting of the Korean Astronomical Society	*Jeonju*
2022.08	**P**	IAU General Assembly	*Busan*
2022.04	**O**	Meeting of the Korean Astronomical Society	*Busan*
2022.01	**O**	East-Asia ALMA Workshop	*Online*
2021.10	**O**	1st KooGiG-Junior Workshop	*Online*
2021.04	**P**	Meeting of the Korean Astronomical Society	*Online*
2020.10	**P**	Meeting of the Korean Astronomical Society	*Online*
2019.12	**O**	Galaxy Evolution Seminar, University of Oxford	*Oxford*
2019.10	**O**	Meeting of the Korean Astronomical Society	*Cheongsong*
2019.04	**P**	Meeting of the Korean Astronomical Society	*Busan*

Computing Skills

Advanced

- Python (numpy, scipy, astropy, SpectralCube, matplotlib, pycprops)
- CASA (ALMA & VLA data reduction, imaging, self-calibration)
- LaTeX / Linux / macOS / Bash
- 2DBAT, 3DBarolo, SoFiA-2

Intermediate

- IDL, MATLAB, Fortran
- AIPS, IRAF, Sextractor

Academic References

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Postdoctoral supervisor (current)

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Prof. Martin Bureau
WISDOM collaboration lead; host of Oxford research visit

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Dr. Chang-Goo Kim

Collaborator on ram pressure stripping related works (TIGRESS and VLA)

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